

Kinetics of the reaction of vinyl radicals with NO: *Ab initio* theory, master equation predictions, and laser absorption measurements†

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The pulsed laser photolysis/cw laser absorption technique is used to investigate the reaction of vinyl (C_2H_3) with NO in the temperature range from 295 to 700 K and pressures from 10 to 320 Torr (1.33 to 42.6 kPa). Vinyl radicals are generated by photolysis of vinyl iodide at 266 nm and detected by visible laser absorption in a vibronic band of the ($\tilde{A} \leftarrow \tilde{X}$) transition near 403 nm. The potential energy surface is explored with both quadratic configuration interaction and multi-reference configuration interaction *ab initio* calculations. These *ab initio* predictions are employed in RRKM theory based master equation simulations of the temperature and pressure dependent kinetics. At room temperature, the overall rate constant for removal of vinyl radical by NO is measured to be $1.6 \pm 0.4 \times 10^{-11} \text{ cm}^3 \text{ molecule}^{-1} \text{ s}^{-1}$, with negligible pressure dependence from 10 Torr (1.33 kPa) to 160 Torr (21.3 kPa) of helium. At constant pressure the rate constant decreases rapidly with temperature. At higher temperatures, a falloff of the rate constant to lower pressure is observed. The *ab initio* characterizations suggest a significant contribution from $HCN + CH_2O$ formation, with both isomerization transition states for the pathway leading to this product lying $\sim 15 \text{ kcal mol}^{-1}$ (63 kJ mol^{-1}) below the entrance channel. The master equation analysis provides a reasonably satisfactory reproduction of the observed kinetic data. The $HCN + CH_2O$ bimolecular channel, which proceeds from the addition complex through tight ring forming and opening transition states, has a negative temperature dependence and is the dominant channel for pressures of about 50 Torr (6.7 kPa) and lower. The theoretically predicted zero pressure rate coefficient is reproduced by the modified Arrhenius expression $5.02 \times 10^{-11} (T/298)^{-3.382} \exp(-516.3/T) \text{ cm}^3 \text{ molecule}^{-1} \text{ s}^{-1}$ (with T in K).

Introduction

The vinyl radical (C_2H_3) is an important intermediate in combustion processes as well as in low temperature extraterrestrial atmospheres. In combustion processes it is a key intermediate arising from the decomposition of higher hydrocarbons in essentially all flames. Furthermore, in certain environments it may play a significant role in the precyclization chemistry leading to soot.¹ NO is known to react rapidly with hydrocarbon radicals and such reactions are thought to suppress soot formation. Furthermore, its reaction with the vinyl radical is central to understanding the pyrolysis of acetylene in NO_2 , and is a key reaction in fuel-rich reburning.

Despite the prominence of the vinyl radical in combustion chemistry, measurements of its reaction kinetics have been relatively scarce. Previous time-resolved experimental studies on the kinetics of vinyl radical reactions have used either time-resolved VUV-absorption³ or mass spectroscopy^{4,5} as a

probe of the vinyl radical concentration. Both techniques suffer from some limitations. Vacuum-ultraviolet absorption is not very specific; often reaction products absorb in the same wavelength region, complicating the analysis of the observed absorption time profiles. Mass spectroscopy, on the other hand, is a very sensitive and specific technique, but experiments are usually restricted to low pressures. Therefore, information on the pressure dependence of radical reactions can be more difficult to obtain. These drawbacks motivate the use of time-resolved visible absorption of the vinyl radical in the $A \leftarrow X$ band^{6–9} as a probe for the investigation of vinyl reactions. The present work describes the application of visible absorption probing to the kinetics of the vinyl + NO reaction.

The reaction of vinyl radical with nitric oxide has been investigated in the context of the mercury-photosensitized decomposition of acetylene/NO mixtures¹⁰ and H- and Cl-initiated reactions of acetylene with NO.¹¹ These experiments indicated that the reaction of vinyl radicals with NO produces $HCN + CH_2O$ at room temperature. Sherwood and Gunning¹⁰ found that the production of HCN in the Hg-sensitized reaction of acetylene/NO mixtures decreased with increasing pressure and could be completely suppressed by sufficiently high buffer gas pressure. In an investigation of acetylene pyrolysis, Benson¹² addressed Sherwood and Gunning's experiments and Ogura's related shock tube experiments,¹³ invoking a

† Electronic supplementary information (ESI) available: Frequencies and geometries for the key stationary points in the $C_2H_3 + NO$ reaction. See <http://www.rsc.org/suppdata/cp/b4/b401163e/>

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chemically activated $\text{C}_2\text{H}_3\text{NO}$ intermediate (presumably nitrosoethene) that could cyclize and dissociate to form $\text{HCN} + \text{CH}_2\text{O}$. He postulated the ring-closure step to form 1,2-oxazete as rate-limiting in the transformation of nitrosoethene to $\text{HCN} + \text{CH}_2\text{O}$ and derived an activation energy for nitrosoethene $\rightarrow \text{HCN} + \text{CH}_2\text{O}$ of $\sim 42.5 \text{ kcal mol}^{-1}$ (178 kJ mol^{-1}) (or $\sim 6 \text{ kcal mol}^{-1}$ (25 kJ mol^{-1}) below vinyl + NO) by analogy with similar ring closure reactions.

Recently Sumathi *et al.*¹⁴ have claimed, based on *ab initio* calculations, that the formation of bimolecular products from vinyl + NO should have a significant activation energy, and that the reaction should proceed by simple addition except at high temperatures. Indeed, their second order Møller–Plesset perturbation theory (MP2) calculations with a 6-311++G(d,p) d,p) basis set suggest a barrier of $8.6 \text{ kcal mol}^{-1}$ (36 kJ mol^{-1}) for the lowest energy bimolecular channel. However, their highest-level calculations, employing coupled cluster theory with singles, doubles, and perturbative triples (CCSD(T)) and the same basis set, suggest that the transition states for the bimolecular channel forming $\text{HCN} + \text{CH}_2\text{O}$ should in fact be lower in energy than the vinyl + NO reactants, with the highest barrier lying slightly ($0.8 \text{ kcal mol}^{-1}$ (3.3 kJ mol^{-1})) below reactants.

In the present work absolute total removal rate coefficients for vinyl radical reacting with nitric oxide are measured as a function of temperature (298–700 K) and pressure (10–320 Torr (1.33–42.6 kPa)). The vinyl radical is produced by pulsed photolysis of vinyl iodide and probed by using long-path visible absorption in the $(\tilde{A} \leftarrow \tilde{X})$ band.^{6–9} The reaction displays a negative temperature dependence and displays pressure falloff above room temperature.

The present work also provides a higher level quantum chemical analysis of the key stationary points on the vinyl + NO potential energy surface. In particular, larger basis sets are employed within both CCSD(T) and quadratic configuration–interaction calculations including single, double, and perturbative triples excitations (QCISD(T)). More significantly, large basis set multi-reference configuration–interaction calculations are used to resolve some discrepancies and uncertainties in the QCISD(T) and CCSD(T) calculations. The net result is a confident prediction that the pathway to forming $\text{H}_2\text{CO} + \text{HCN}$ has a maximum barrier that lies well below the reactants ($\sim 14 \text{ kcal mol}^{-1}$ (59 kJ mol^{-1}) below).

A final component of this work involves an RRKM theory based master equation modeling of the temperature and pressure dependence of the kinetics. These master equation simulations demonstrate that the observed rate coefficients for vinyl radical removal are consistent with a transformation from predominantly addition at the highest pressures studied experimentally to predominantly bimolecular reaction at the lowest pressures studied.

Experiment

The technique of laser photolysis/long path absorption is used to measure the kinetics of the vinyl radical reaction with NO. Vinyl radicals are formed by pulsed laser photolysis and the decay of the vinyl radical concentration is monitored by absorption at either the $\sim 403 \text{ nm}$ or the $\sim 425 \text{ nm}$ band of the $(\tilde{A} \leftarrow \tilde{X})$ transition.^{6,9} The doubled output of a ring Ti:Al₂O₃ laser serves as the probe laser. Part of the probe is split off using an uncoated fused silica flat and sent to a detector (I_0). The remainder of the probe beam is passed multiple times through the reactor using a Herriott-type geometry,¹⁵ and directed to a second detector (I). A variable neutral density wheel is used before one detector to balance the power reaching the two detectors in the absence of vinyl radical. The absorption in the reactor is then monitored as the difference of the two balanced detectors ($I - I_0$). The change in

absorption upon photolysis is small ($\Delta I/I_0 < 1\%$) and is treated as linearly proportional to vinyl concentration.

In the present experiments scattered ultraviolet photolysis light can produce transient changes in the multipass mirror reflectivity, presumably *via* light absorption in either the dielectric coating or the substrate. This transient response, although small, can be a significant obstacle to measuring time-resolved kinetics by visible absorption, because the change in transmission is of similar magnitude to the desired signal. However, this artifact is reduced at longer photolysis wavelength; at 266 nm, it can be virtually eliminated. This consideration restricts the choice of photolytic sources, and significant effort has been expended searching for suitable vinyl sources at 266 nm. In the present experiment vinyl radicals are formed by 266 nm photolysis of vinyl iodide, which is expected to produce both vinyl + I and acetylene + HI products. Dissociation of $\text{C}_2\text{H}_3\text{I}$ at 254 nm produces vinyl + I in about 83% yield.¹⁶ The presence of C_2H_2 and HI should have a negligible effect on the present results. The reaction of vinyl with acetylene is very slow,^{5,17} but the reaction of vinyl with HI could have a significant rate constant.^{18,19} The reaction of vinyl with either of the unwanted photoproducts will not vary with NO concentration, but will contribute to the second-order component of the decay observed at zero NO concentration. The pseudo-first order reaction with NO is much larger than the second-order reaction under the conditions of these experiments.

The photolysis laser passes along the axis of the Herriott resonator and intersects the probe beam in the center of the reactor, where the temperature is well-controlled. The cell is heated by a set of three resistive heaters that operate under microprocessor control, with the temperature monitored by three K-type thermocouples inside the reactor. The reactants are delivered to the cell by individual calibrated mass flow controllers and the overall pressure in the cell is controlled by a butterfly valve in the pump. Nitric oxide (99.9%) is passed through a cold trap held at 193 K (dry ice–acetone) to remove NO₂ impurities. The vinyl iodide is subjected to several freeze–pump–thaw cycles. According to the manufacturer's lot analysis, the vinyl iodide (technical grade, 85%) contains oxolane (tetrahydrofuran, THF) as the principal impurity. The absorption of THF at 266 nm is negligible, and the impurity is not expected to affect the present measurements. Vinyl iodide samples containing unacceptable levels of molecular iodine impurity are rinsed with sodium bisulfite, filtered, and dried over magnesium sulfate.

A typical transient absorption trace is shown in Fig. 1. The observed data traces are fit to single exponential decays. This analysis assumes that the contribution of the vinyl–vinyl reaction and other second-order removal of vinyl radical can be neglected. Under the present experimental conditions, the pseudo-first order removal by reaction with NO dominates, as seen from the semi-logarithmic plot of absorption *vs.* time shown in the inset of Fig. 1. The reported rate coefficients are derived from a linear least-squares fit to a plot of inverse decay times *versus* NO concentration, as shown in Fig. 2, weighted by the inverse of the statistical uncertainties in the individual decay constant determinations. The slope of this plot yields the second-order rate coefficient for the reaction of vinyl with NO. The zero intercept reflects the removal of vinyl radical by processes that do not depend on the NO concentration, including vinyl radical self-reaction, diffusion out of the probe volume, and possible reaction with other products of the photolysis. Several (3–5) rate coefficient determinations are typically carried out at each temperature and pressure, and the reported value is the mean of these determinations, weighted by the inverse of their statistical uncertainties. The final overall uncertainty bound given is a convolution of the statistical uncertainties (2σ) and estimates of possible systematic inaccuracies (*e.g.*, in concentration determinations).

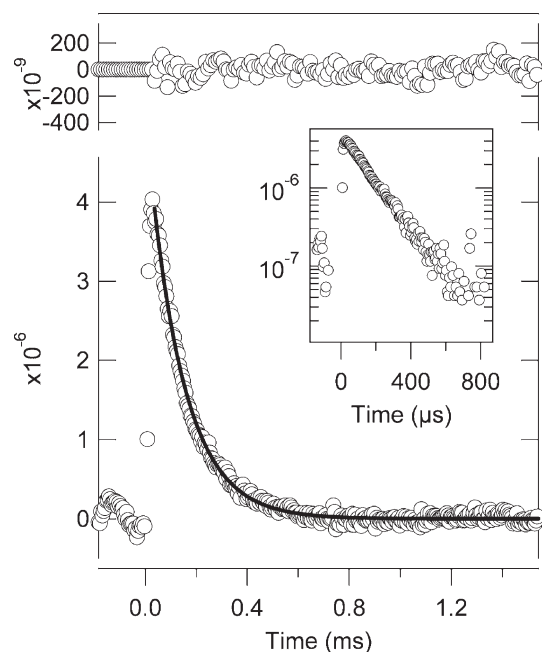


Fig. 1 Time-resolved absorption signal (open circles) following the photolysis of $\text{C}_2\text{H}_3\text{I}$ in the presence of $[\text{NO}] = 2.32 \times 10^{15} \text{ cm}^{-3}$. For clarity, every fifth data point is shown. The decay of the absorption is fit to a single exponential, shown by the solid line. Residuals to the fit ($\times 5$) are given above.

Theory

Quantum chemistry

The geometric structures and vibrational frequencies for all stationary points considered here are obtained *via* density functional theory employing the Becke-3 Lee–Yang–Parr (B3LYP) functional²⁰ and the 6-311++G(d,p) basis set.²¹ Spin restricted wavefunctions are generally employed. Zero-point energy corrections are evaluated at the B3LYP/6-311++G(d,p) level and are incorporated in all the final energies reported below. The present density functional calculations have all been performed with the Gaussian98 quantum chemistry program.²²

Quadratic configuration–interaction calculations with perturbative inclusion of the triples contribution, QCISD(T),²³ are employed to obtain higher-level estimates E_{HL} at these

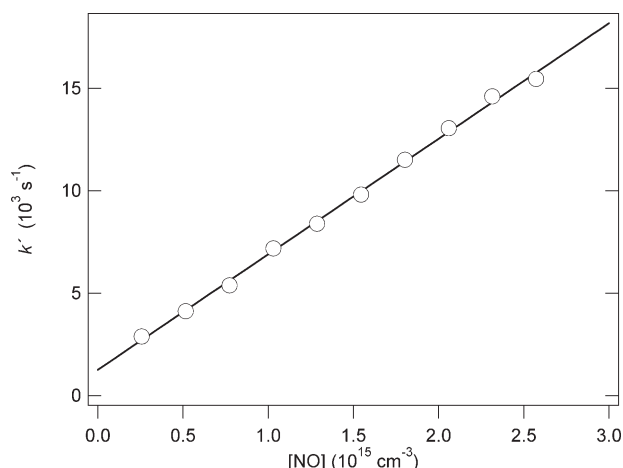


Fig. 2 Plot of the fitted pseudo-first order rate constant (open circles) versus NO concentration, at 600 K and 80 Torr (10.6 kPa) total pressure. The second-order rate constant is obtained from a linear fit to the data, shown as the solid line, weighted by the errors in the individual determinations.

stationary points.²⁴ These calculations employ Dunning's correlation consistent polarized valence triple- (cc-pvtz) and quadruple-zeta (cc-pvqz) basis sets.²⁵ The infinite basis set limit is estimated *via* the extrapolation^{24,26}

$$E(\infty) = E(l_{\text{max}}) - B/(l_{\text{max}} + 1)^4 \quad (1)$$

where l_{max} is the maximum angular momentum in the basis set. For the cc-pvtz and cc-pvqz pair, with $l_{\text{max}} = 3$ and 4, respectively, this extrapolation yields

$$E_{\text{HL}} = E_{\text{QCISD(T)/cc-pvqz}} + 0.6938 \\ \times (E_{\text{QCISD(T)/cc-pvqz}} - E_{\text{QCISD(T)/cc-pvtz}}) \quad (2)$$

These energies are converted to absolute 0 K heats of formation *via* comparison with related calculations and experimental values for H_2 , CH_4 , CO_2 , and N_2 . Spin-restricted wavefunctions are used for all the QCISD(T) and CCSD(T) calculations.

For the ring opening transition state (cyclic- $\text{OCH}_2\text{CHN} \rightarrow \text{OCH}_2 + \text{HCN}$) both spin restricted (R) and spin unrestricted (U) B3LYP density functionals are employed. The two solutions yield markedly different energies and structures, with the U-B3LYP saddlepoint lying 20 kcal mol^{-1} (84 kJ mol^{-1}) below the R-B3LYP saddlepoint, and having a major spin contamination corresponding to an average spin-squared of 0.80. The optimized geometries, which are illustrated in Fig. 3, indicate that the R-B3LYP saddlepoint lies much closer to products, with a CC bond length that is $0.09 \text{ \AA}$ (9 pm) greater, a CO bond length that is $0.08 \text{ \AA}$ (8 pm) smaller, and an OCCN torsion that differs significantly from 0. The R-B3LYP geometry corresponds quite closely to the MP2/6-311++G(d,p) geometry of Peeters and coworkers.¹⁴

Spin restricted QCISD(T) calculations at these two predicted geometries for the ring opening saddlepoint also yield markedly different energies, with that at the U-B3LYP geometry being 10 kcal mol^{-1} (42 kJ mol^{-1}) lower. Notably, the T1 diagnostic^{27,28} for the R-B3LYP geometry is extraordinarily large (0.17) indicating major uncertainties in the ability of the QCISD(T) method to treat this saddlepoint. Interestingly, at the U-B3LYP geometry this diagnostic is smaller (0.091), but still large enough to cause significant concern. To explore these difficulties further, spin restricted CCSD(T) calculations

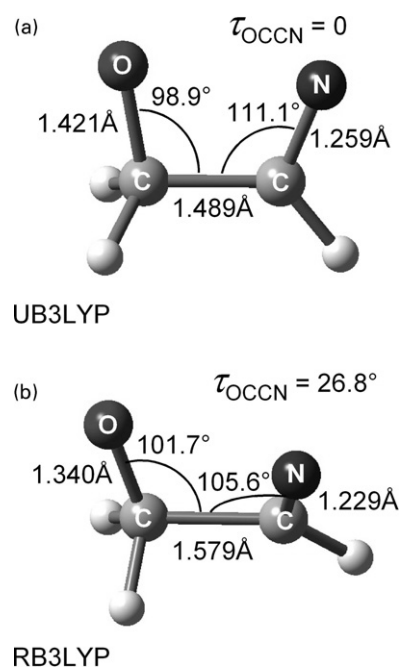


Fig. 3 U-B3LYP/6-311++G(d,p) and R-B3LYP/6-311++G(d,p) predicted geometries for the stationary point connecting nitrosoethene to 1,2-oxazete. $1 \text{ \AA} = 100 \text{ pm}$.

are also performed at these geometries. At the R-B3LYP geometry, the CCSD(T)/6-311G(d,p) result is 16 kcal mol⁻¹ (67 kJ mol⁻¹) below the corresponding QCISD(T)/6-311G(d,p) energy, and for the UB3LYP geometry this difference is 10 kcal mol⁻¹ (42 kJ mol⁻¹). This large difference between QCISD(T) and CCSD(T) predictions (typically QCISD(T) and CCSD(T) predictions differ by less than 0.2 kcal mol⁻¹) is a further indication of the unreliability of single-reference based treatments for this saddlepoint. At the U-B3LYP geometry the difference between the QCISD(T) and CCSD(T) predictions is reduced somewhat, but is still 7 kcal mol⁻¹ (30 kJ mol⁻¹).

Intrinsic reaction path evaluations have been performed from the UB3LYP saddlepoint. Higher level evaluations along this reaction path should help to clarify the relative importance of the two different predictions for the geometry of the saddlepoint for opening of the OCCN ring. Unfortunately, the QCISD(T) calculations failed to converge for moderate displacements along the UB3LYP reaction path. Furthermore, these calculations suggest a significant displacement of the QCISD(T) saddlepoint from the UB3LYP one, with the convergence failures arising prior to reaching the maximum in the QCISD(T)/UB3LYP energies. In any case, the large differences between the QCISD(T) and CCSD(T) predictions would still yield great uncertainties in any such QCISD(T)/B3LYP predictions.

These uncertainties led us to perform internally-contracted multi-reference configuration–interaction calculations.^{29,30} These calculations are performed with complete-active-space (CAS) self-consistent-field (SCF) reference wavefunctions and include all single and double excitations from the CAS reference space (CAS + 1 + 2). The incorporation of the Davidson correction for higher order excitations is denoted here by CAS + 1 + 2 + Q. The cc-pvtz and cc-pvqz basis sets are again used to extrapolate to the infinite basis set limit as in eqn. (2). A 10-electron, 8-orbital active space is used for the CAS calculations. For the products (HCN + CH₂O) these orbitals correlate with the CN and CO π and π^* orbitals, the O lone pair orbital, and the A'' out-of-plane CH₂ σ orbital. For the reactants (cyclic-OCH₂CHN) these orbitals correlate with CC, CO, and NO σ and σ^* orbitals, the O lone pair orbital, and the A'' out-of-plane CH₂ σ orbital.

For completeness, related U-B3LYP/6-311++G(d,p) and CAS + 1 + 2 calculations are performed for the ring closing transition state (*cis*-CH₂CHNO → cyclic-OCH₂CHN). In this instance, the unrestricted solution reduced to the restricted solution with no spin-contamination. The CAS reference space for this CAS + 1 + 2 calculation consisted of all the occupied valence orbitals plus the 2 virtual orbitals that correlate with the CN π^* and CO σ^* orbitals in the ring. In *cis*-CH₂CHNO these virtual orbitals correlate with the CC and NO π^* orbitals. The B3LYP/6-311++G(d,p) intrinsic reaction path is also obtained. However, for this reaction path, the maximum in the CAS + 1 + 2 energies along the B3LYP reaction path is essentially identical to the value at the B3LYP saddlepoint, and so only the saddlepoint energies are reported below.

Finally, a few additional B3LYP, QCISD(T), and CAS + 1 + 2 calculations are performed in order to more definitively rule out the channels involving H transfers and to examine the possible importance of a bicyclic ring forming channel. The CAS reference space for the CAS + 1 + 2 calculations on the bicyclic ring formation pathway includes 8 electrons in 8 active orbitals. These active orbitals correlate with the CC and NO σ , σ^* , π^* orbitals in *trans*-nitrosoethene.

The present QCISD(T), CCSD(T) and CAS + 1 + 2 calculations are performed with the MOLPRO quantum chemistry package³¹ on a Linux cluster.³²

Kinetics

The methodology employed in the present master equation simulations is as described elsewhere.^{33–36} Briefly, the *ab initio*

predictions for the saddlepoint properties are implemented in RRKM theory analyses for the individual channel microcanonical rate constants. These analyses employ rigid-rotor harmonic-oscillator assumptions for all but the CCNO torsional mode in the entrance adduct, which is treated as a hindered internal rotor, and in the estimates for the entrance channel transition state, where the relative rotational modes are also treated as hindered internal rotors employing the variable reaction coordinate transition state theory formalism.^{37,38} The torsion and relative rotations are the key modes to be treated anharmonically; negligible errors are made by a harmonic approximation for the remaining modes.

For the entrance channel, the radical nature of the two reactants implies that the addition process is likely barrierless. The quantitatively accurate *a priori* prediction of rate coefficients for such reactions requires quantitative estimates of the potential over wide ranges of geometry. Obtaining such information is difficult and time-consuming and is beyond the scope of this study. Instead, a simple model potential is developed and incorporated in variable reaction coordinate transition state theory evaluations to evaluate the entrance channel flux. The particular potential form employed is closely analogous to that used in related studies of the C₂H₅ + O₂ reaction.^{33,39,40} The currently employed parameters are designed to provide a realistic 300 K high pressure addition rate of 2.2×10^{-11} cm³ molecule⁻¹ s⁻¹; this particular value is chosen primarily on the basis of the experimentally observed rate coefficient at room temperature. The calculated high pressure limit falls by approximately 15% between 300 and 700 K.

The master equation calculations are carried out using the VARIFLEX program.⁴¹ The zero-pressure predictions are performed with the 2-dimensional master equation in energy E and total angular momentum J .³⁶ The pressure-dependent evaluations employ a reduction to the one-dimensional master equation in E . In the low-pressure limit the one-dimensional and two-dimensional solutions are found to differ by about 10% or less. The pressure-dependent evaluations employ an average downwards energy transfer parameter, $\langle \Delta E_{\text{down}} \rangle$, of 200 (T (K)/298) cm⁻¹. This form, which is found to provide a good fit to the present experimental data, is consistent with related findings for a number of other reactions,^{42–44} and, at room temperature, with the numerical value employed in our study of the C₂H₅ + O₂ reaction.^{33,39,40}

Results and discussion

The experimental rate coefficients for the reaction of vinyl radical with NO are shown in Fig. 4 and reported in Table 1. The reaction displays a negligible pressure dependence at 295 K, but the rate coefficient exhibits a clear pressure falloff at higher temperatures. At constant pressure the rate coefficient decreases quite rapidly with temperature.

Energies and frequencies for many stationary points on the vinyl-NO surface have been calculated by Sumathi *et al.*¹⁴ Based on their report, the kinetically significant intermediate species in the vinyl + NO reaction can be identified as *cis*- and *trans*-nitrosoethene (the two conformers of the initial C₂H₃NO addition complex), and 1,2-oxazete (the four-membered heterocycle that leads to HCN + CH₂O formation). These species, and the transition states linking them to reactants and products, are studied in the present work at the QCISD(T) level and at the CAS + 1 + 2 level, with both calculations including an extrapolation to the infinite basis set limit. The energies for the stationary points are listed in Table 2, with the optimal estimates illustrated in Fig. 5. Frequencies and geometries are supplied as Electronic supplementary information (ESI).†

The CAS + 1 + 2 + Q calculations should provide the best estimate for the heats of formation of the ring closing and

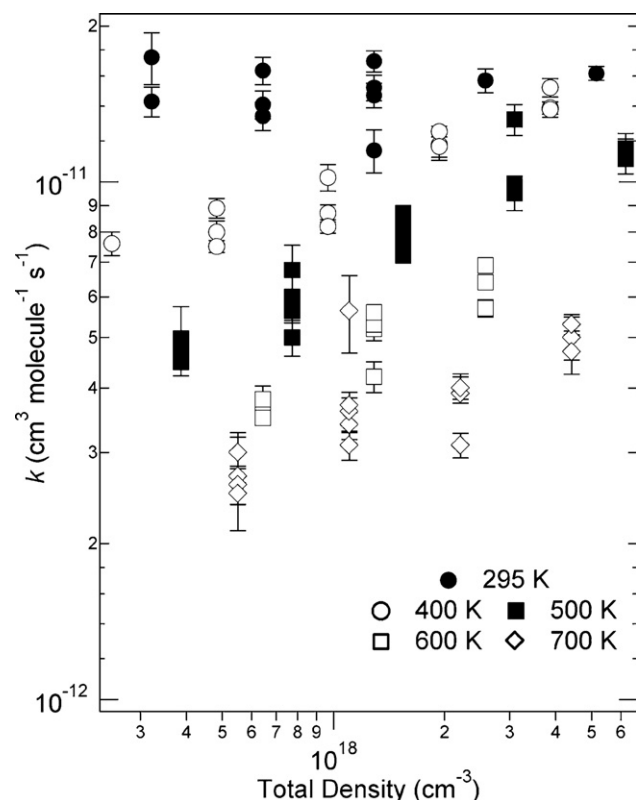


Fig. 4 Plot of observed rate coefficients for vinyl radical removal by NO versus total density for a range of temperatures. The error bars represent the $\pm 2\sigma$ precision of the individual determinations. The values plotted in Figs. 6 and 7 and reported in Table 1 are the weighted mean of the individual determinations shown here.

opening saddlepoints. Meanwhile the extrapolated QCISD(T) energies should provide the best estimates for the remaining heats of formation because none of them have an exceedingly large T1 diagnostic. The values of $79.7 \text{ kcal mol}^{-1}$ ($333.5 \text{ kJ mol}^{-1}$) and $79.5 \text{ kcal mol}^{-1}$ ($332.6 \text{ kJ mol}^{-1}$) for the ring closing and opening saddlepoints places them 14.6 and $14.8 \text{ kcal mol}^{-1}$ below reactants, respectively. The evaluations along

Table 1 Experimentally measured rate coefficients for the reaction of C_2H_3 with NO. The uncertainty estimates are $\pm 2\sigma$

Temperature	Pressure	$k/10^{-12} \text{ cm}^3 \text{ molecule}^{-1} \text{ s}^{-1}$
295 K	10 Torr (1.33 kPa)	(15 ± 3)
	20 Torr (2.66 kPa)	(15 ± 2)
	40 Torr (5.32 kPa)	(15 ± 2)
	80 Torr (10.6 kPa)	(16 ± 2)
	160 Torr (21.3 kPa)	(18 ± 2)
400 K	10.4 Torr (1.38 kPa)	(7.6 ± 0.8)
	20 Torr (2.66 kPa)	(8.0 ± 0.7)
	40 Torr (5.32 kPa)	(8.8 ± 0.8)
	80 Torr (10.6 kPa)	(12 ± 1)
	160 Torr (21.3 kPa)	(14 ± 1)
500 K	20 Torr (2.66 kPa)	(4.7 ± 0.9)
	40 Torr (5.32 kPa)	(5.8 ± 0.9)
	80 Torr (10.6 kPa)	(7.8 ± 0.6)
	160 Torr (21.3 kPa)	(10.5 ± 1.4)
	320 Torr (42.6 kPa)	(11.4 ± 1.5)
600 K	40 Torr (5.32 kPa)	(3.6 ± 0.4)
	80 Torr (10.6 kPa)	(5.4 ± 0.4)
	160 Torr (21.3 kPa)	(6.4 ± 0.4)
700 K	40 Torr (5.32 kPa)	(2.6 ± 0.7)
	80 Torr (10.6 kPa)	(3.5 ± 0.5)
	160 Torr (21.3 kPa)	(3.8 ± 0.4)
	320 Torr (42.6 kPa)	(5.1 ± 0.8)

the U-B3LYP/6-311++G(d,p) reaction path for the ring opening yield a clear maximum in the CAS + 1 + 2 + Q extrapolated basis set curve. This maximum is expected to provide a reasonable approximation to that which would have been obtained from optimizations at the CAS + 1 + 2 + Q level. Its use might provide somewhat of an overestimate of the CAS + 1 + 2 + Q saddlepoint energy, but should not significantly underestimate it. The variations between the inclusion and neglect of the Davidson correction and between the referencing to the reactants and products in the two isomerizations suggest that the uncertainties in these estimates are only about 2 to 4 kcal mol^{-1} (8 – 17 kJ mol^{-1}). Thus, it can be confidently concluded that the pathway from reactants to bimolecular products has a maximum barrier that is well below reactants. As a result, at low temperature the reaction should proceed with a rate coefficient that is equal to the high-pressure addition rate, independent of pressure.

The entropies for the ring-closing and ring-opening transition states are much lower than for the entrance channel. Therefore, at higher temperatures one expects the rate of decomposition of the initial adduct back to reactants to exceed the rate of production of bimolecular products from it. As a result, one of the ring transition states will become the dominant rate-limiting transition state at higher temperatures. Thus, in the high-temperature limit one expects the rate of loss of reactants to be limited by the zero-pressure limit rate coefficient, independent of pressure.

The experimental results indicate some pressure dependence to the kinetics in the 300 to 700 K region and hence appear to be in an intermediate pressure regime. These results provide a valuable test for the proposed theoretical model. A comparison of the master equation predictions and the experimental results is provided in Fig. 6. The agreement between the full master equation model and the experimental observations is quite good, and is likely within reasonable combined theoretical and experimental uncertainties (the maximum discrepancy is less than 30%). This agreement validates the basic theoretical model, particularly the presence of a maximum barrier for bimolecular reaction that is well below reactants.

For the purposes of illustration, we provide in Fig. 7 the theoretical predictions obtained when considering only the entrance channel. These results clearly fall off too quickly with decreasing pressure. One can slightly improve the agreement for this “addition-only” case via variations in $\langle \Delta E_{\text{down}} \rangle$ and the high-pressure limit rate coefficient. However, such fits still deviate significantly from the experimental observations at the lowest pressures. Furthermore, the fitted $\langle \Delta E_{\text{down}} \rangle$ values are at least twice as high, probably unphysically large. The energy transfer parameter is sometimes used as a “fitting parameter,” but recent investigations show that similar reaction systems exhibit similar energy transfer parameters when used in a consistent set of master equation calculations.⁴³

The present theoretical predictions for the branching ratio to form $\text{HCN} + \text{CH}_2\text{O}$ are illustrated in Fig. 8. Notably, the present experimental observations span the region of the transition from primarily $\text{HCN} + \text{CH}_2\text{O}$ products at low pressure to primarily stabilized nitrosoethene product at high pressure. The theoretical predictions for the branching ratio are in qualitative but not quantitative accord with the observations of Sherwood and Gunning,^{10,11} who measured a decrease of the HCN yield from Hg-sensitized C_2H_2 reaction with NO as the buffer gas pressure increased. At room temperature, Sherwood and Gunning¹⁰ report a reaction yield of HCN as ~ 0.16 at 10 Torr (1.33 kPa) and ~ 0.07 at 20 Torr (2.66 kPa) cell pressure. In contrast, the present calculated bimolecular product yield drops from 0.94 at 10 Torr to 0.81 at 20 Torr .

The calculated zero-pressure rate coefficient for the production of $\text{HCN} + \text{CH}_2\text{O}$ as a function of temperature is $8.3 \times 10^{-12} \text{ cm}^3 \text{ molecule}^{-1} \text{ s}^{-1}$ at 298 K and falls to $1.3 \times 10^{-12} \text{ cm}^3 \text{ molecule}^{-1} \text{ s}^{-1}$ at 700 K . The negative temperature

Table 2 Quantum chemical heats of formation (0 K) for the key stationary points in the $C_2H_3 + NO$ reaction^a

Species	B3LYP ^b	QCISD(T) ^c	T1 ^d	CAS + 1 + 2 ^e	CAS + 1 + 2 + Q ^f
C_2H_3	69.9 (292.5)	72.0 (301.2)	0.021	—	—
NO	18.9 (79.1)	22.3 (93.3)	0.024	—	—
$C_2H_3 + NO$	88.8 (371.5)	94.3 (394.6)	—	—	—
<i>trans</i> -CH ₂ CHNO	40.5 (169.5)	42.3 (177.0)	0.016	—	—
<i>cis</i> -CH ₂ CHNO	44.8 (187.4)	45.9 (192.0)	0.017	—	—
ts2 (<i>cis</i> -CH ₂ CHNO ↔ cyclic-CH ₂ CHNO)	79.1 (330.9)	78.2 (327.2)	0.041	83.0, 78.4 80.7 (337.6)	80.5, 78.9 79.7 (333.5)
cyclic-CH ₂ CHNO	46.2 (193.3)	43.7 (182.8)	0.018	—	—
ts3 (cyclic-CH ₂ CHNO ↔ CH ₂ O + HCN) ^g	93.1 (389.5)	85.3 (356.9)	0.165	—	—
ts3 (cyclic-CH ₂ CHNO ↔ CH ₂ O + HCN) ^{h,i}	73.6 (307.9)	75.5 (315.9)	0.091	73.4, 82.7 78.1 (326.8)	75.1, 78.9 77.0 (322.2)
ts3 (cyclic-CH ₂ CHNO ↔ CH ₂ O + HCN) ^{h,j}	—	—	—	75.2, 84.5 79.9 (334.3)	77.6, 81.4 79.5 (332.6)
CH ₂ O	−25.5 (−106.7)	−25.3 (−105.9)	0.017	—	—
HCN	30.5 (127.6)	31.0 (129.7)	0.015	—	—
CH ₂ O + HCN	5.0 (20.9)	5.7 (23.8)	—	—	—

^a Energies are in kcal mol^{−1} (kJ mol^{−1}) and include B3LYP/6-311++G(d,p) zero-point corrections. ^b B3LYP/6-311++G(d,p) energies. ^c QCISD(T)/B3LYP/6-311++G(d,p) energies extrapolated to the infinite basis set limit according to eqn. (2). ^d T1 diagnostic for the QCISD(T)/cc-pvqz calculations. ^e CAS + 1 + 2//B3LYP/6-311++G(d,p) energies extrapolated to the infinite basis set limit according to eqn. (2). The first entry is obtained relative to nitrosoethene, and the second entry (in *italics*) is obtained relative to the 1,2-oxazete (*i.e.*, the cyclic form). The entry in bold is the average of these two entries. ^f As in footnote e, but including the Davidson correction for higher order excitations. ^g Employing a spin restricted wavefunction in the B3LYP geometry determination. ^h Employing an unrestricted spin wavefunction in the B3LYP geometry determination. ⁱ At the B3LYP/6-311++G(d,p) saddlepoint. ^j At the maximum along the B3LYP/6-311++G(d,p) reaction path for the stated method.

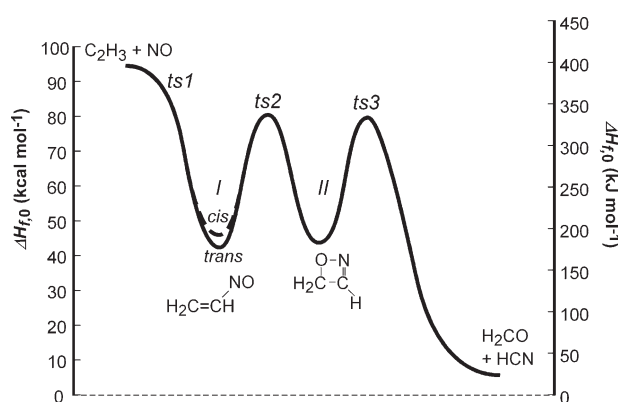


Fig. 5 Relevant stationary point energies on the C_2H_3NO surface, calculated at the QCISD(T) level with a cc-pvtz and cc-pvqz based extrapolation to the infinite basis set limit and B3LYP/6-311++G(d,p) zero-point energy corrections. Energies for transition states ts2 and ts3 are CAS + 1 + 2 + Q values.

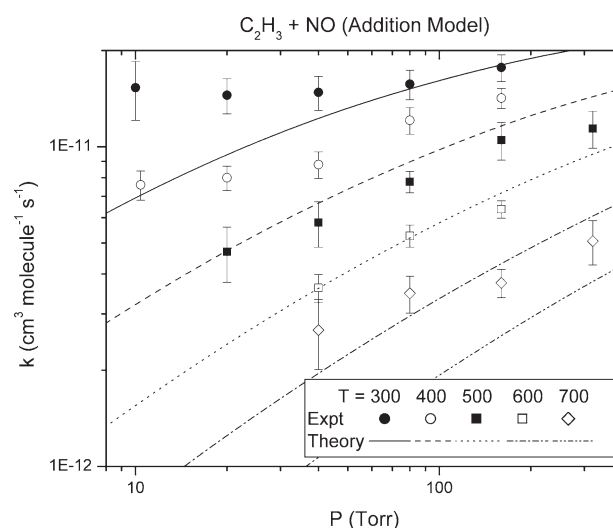


Fig. 7 Comparison of the experimental results (symbols) with master equation calculations assuming only a simple addition reaction (lines). 1 Torr = 133 Pa.

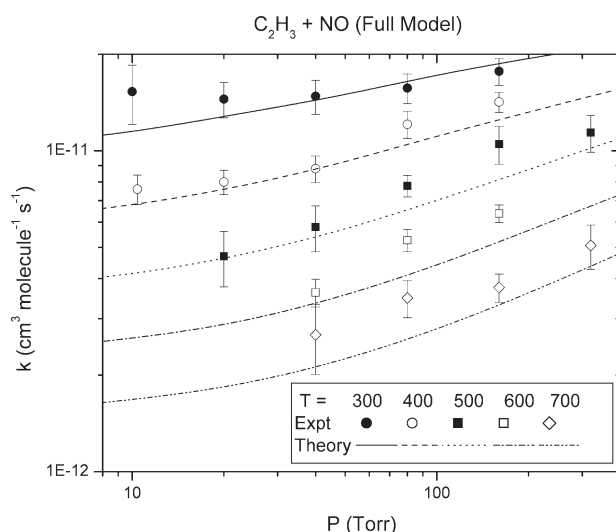


Fig. 6 Comparison of the experimental results (symbols) with master equation calculations including the calculated HCN + CH₂O channel (lines). 1 Torr = 133 Pa.

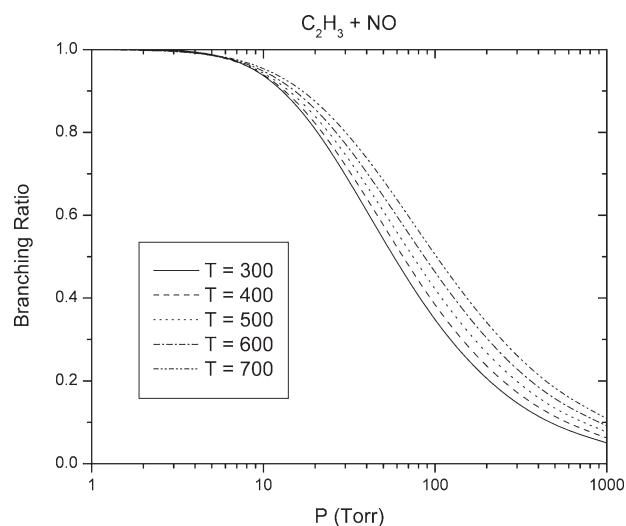


Fig. 8 Plot of the predicted branching ratio for forming HCN + CH₂O versus pressure for temperatures ranging from 300 to 700 K. 1 Torr = 133 Pa.

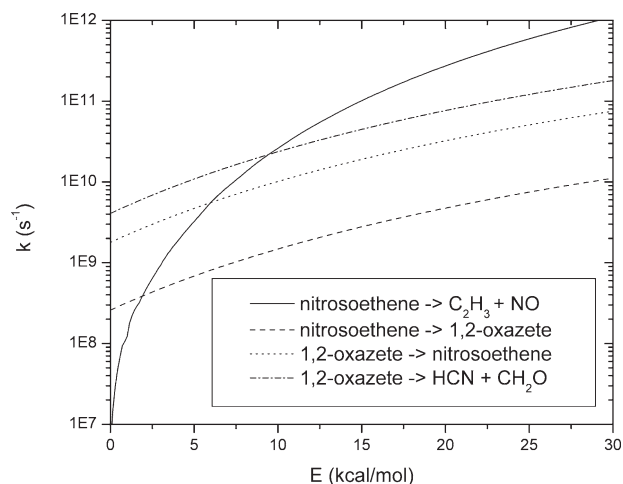


Fig. 9 Plot of the microcanonical rate coefficients for decomposition of nitrosoethene and 1,2-oxazete *versus* energy relative to $\text{C}_2\text{H}_3 + \text{NO}$. 1 kcal = 4.184 kJ.

dependence reflects the dominance of entropic factors as the reaction proceeds through two sterically-constrained transition states. In particular, it is due to the fact that at higher temperatures back-dissociation *via* the loose entrance channel transition state is favored over isomerization to the 4 membered ring and on to products *via* the tight transition states. The predicted zero-pressure values for temperatures ranging from 200 to 2000 K are well fit by the modified Arrhenius expression $5.02 \times 10^{-11} (T/298)^{-3.382} \exp(-516.3/T) \text{ cm}^3 \text{ molecule}^{-1} \text{ s}^{-1}$, with T in K.

The energy dependence of the isomerization and decomposition rates both from nitrosoethene and from 1,2-oxazete are plotted in Fig. 9 for a total angular momentum $J = 2$ (only minor qualitatively insignificant differences are expected for other angular momentum values). At low energies, the slowest reaction is the decomposition back to reactants from nitrosoethene. Then, as the energy increases, the isomerization from nitrosoethene to 1,2-oxazete gradually becomes a comparable bottleneck to the decomposition, and ultimately becomes the

dominant rate limiting step at an excess energy relative to reactants of $2.0 \text{ kcal mol}^{-1}$ (8 kJ mol^{-1}). Note that the rate of isomerization of 1,2-oxazete back to nitrosoethene never competes with the rate of bimolecular product formation from 1,2-oxazete and so the ring opening transition state never provides a significant bottleneck. This observation of the importance of the ring closure step is in accord with Benson's earlier qualitative predictions,¹² although the present prediction for the activation energy is about 5 to 10 kcal mol^{-1} (20 to 40 kJ mol^{-1}) lower than his estimate.

The uncertainty in the energies from Sumathi *et al.*,¹⁴ as evidenced by our greatly different predictions for the ring opening saddlepoint, is quite large. This large difference suggests that the pathway involving an internal H transfer from CH_2CHNO to CH_2CNOH might also be significant. The energy for this saddlepoint is also evaluated here at the QCISD(T) level with an extrapolation to the infinite basis set limit, and is reported in Table 3. The moderate value of 0.030 for the T1 diagnostic at this saddlepoint suggests that the predicted barrier of 9 kcal mol^{-1} (38 kJ mol^{-1}) relative to reactants should be reasonably reliable. This value for the barrier is high enough that this H transfer channel may be safely neglected, especially given that this transition state also has a fairly low entropy. Similarly, the barrier for H transfer to form CH_3CNO , at 12 kcal mol^{-1} (49 kJ mol^{-1}), is also too high for it to contribute significantly.

During the course of this work it was also realized that a bicyclic intermediate (with CCN and CNO rings) exists and may provide an alternative low energy pathway to bimolecular products. B3LYP and QCISD(T) calculations for the key stationary points in this pathway are also reported in Table 3. There are two routes to forming the bicyclic ring; either the CNO ring or the CNC ring can form first. The latter process is predicted to have a barrier of 11 kcal mol^{-1} (48 kJ mol^{-1}) and should not contribute. The formation of the CNO ring has a lower barrier but the calculations are again plagued by spin contamination and high T1 diagnostic problems. Thus, MRCI calculations were performed for the first step in this process and are also reported in Table 3. These results suggest that the barrier to the CNO ring closure is about 1 kcal mol^{-1} (4 kJ mol^{-1}) above reactants (at least on the RB3LYP pathway). This modest barrier, coupled with the low entropy of

Table 3 Quantum chemical heats of formation (0 K) for additional stationary points in the $\text{C}_2\text{H}_3 + \text{NO}$ reaction^a

Species	B3LYP ^b	QCISD(T) ^c	T1 ^d	CAS + 1 + 2 ^e	CAS + 1 + 2 + Q ^f
<i>trans</i> - $\text{CH}_2\text{CHNO} \leftrightarrow \text{CH}_2\text{CNOH}$	100.8 (421.7)	103.3 (432.2)	0.030	—	—
<i>trans</i> - $\text{CH}_2\text{CHNO} \leftrightarrow \text{CH}_3\text{CNO}$	99.7 (417.1)	106.1 (443.9)	0.037	—	—
<i>trans</i> - $\text{CH}_2\text{CHNO} \leftrightarrow \text{CH}_2\text{CNOH}$	100.3 (419.7)	103.3 (432.2)	0.030	—	—
<i>trans</i> - $\text{CH}_2\text{CHNO} \leftrightarrow \text{CH}_2\text{CHNO}$ (initially the CNO ring, but goes directly to both rings)	100.3 (419.7)	—	0.031	102.0, 98.0 100.0 (418.4)	95.2, 95.3 95.2 (398.3)
CH_2CHNO (CNC and CNO rings)	71.8 (300.4)	67.8 (283.7)	0.015	72.2 (302.1)	67.6 (282.8)
<i>trans</i> - $\text{CH}_2\text{CHNO} \leftrightarrow \text{CH}_2\text{CHNO}$ (CNO ring) ^g	88.9 (372.0)	—	0.132	107.3, 103.4 105.4 (441.0)	102.2, 102.8 102.5 (428.9)
CH_2CHNO (CNO ring) ^g	86.6 (362.3)	92.7 (387.9)	0.069	—	—
CH_2CHNO (CNO ring) $\leftrightarrow$ CH_2CHNO (CNO and CNC rings) ^g	90.9 (380.3)	109.3 (457.3)	0.178	—	—
<i>trans</i> - $\text{CH}_2\text{CHNO} \leftrightarrow \text{CH}_2\text{CHNO}$ (initially the CNC ring; but goes directly to both rings)	106.9 (447.3)	105.7 (442.2)	0.037	—	—
CH_2CHNO (CNC and CNO rings) $\leftrightarrow$ <i>cis</i> - CH_2NCHO	81.6 (341.4)	78.3 (327.6)	0.028	—	—

^a Energies are in kcal mol^{-1} (kJ mol^{-1}) and include B3LYP/6-311++G(d,p) zero-point corrections. ^b B3LYP/6-311++G(d,p) energies.

^c QCISD(T)/B3LYP/6-311++G(d,p) energies extrapolated to the infinite basis set limit according to eqn. (2). ^d T1 diagnostic for the QCISD(T)/cc-pvqz calculations. ^e CAS + 1 + 2/B3LYP/6-311++G(d,p) energies extrapolated to the infinite basis set limit according to eqn. (2). The first entry is obtained relative to *trans*-nitrosoethene, and the second entry (in italics) is obtained relative to the bicyclic form. The entry in bold is the average of these two entries. ^f As in footnote e, but including the Davidson correction for higher order excitations. ^g Employing a spin unrestricted wavefunction in the B3LYP geometry determination. All others employ spin restricted wavefunctions.

the transition state, suggests that this pathway will also be fairly insignificant.

Concluding remarks

In conclusion, the proposed mechanism for the vinyl + NO reaction is a dominant addition channel with a tight transition state to bimolecular products located well below the vinyl + NO asymptote. This mechanism agrees with the present measurements of removal rate coefficients for vinyl radical with NO and qualitatively, but not quantitatively, the literature determinations of product yields from sensitized acetylene-NO reactions. The present work predicts increased production of bimolecular products. Work to characterize product branching ratios as a function of temperature and pressure is presently underway and should help to refine the understanding of this reaction.

The highest-energy transition states for the lowest energy pathway to formation of bimolecular products, that for 4 membered ring-closing from the initially-formed nitrosoethene to 1,2-oxazete and the dissociation of 1,2-oxazete into HCN and CH₂O, are calculated to lie 14.6 kcal mol⁻¹ (61.1 kJ mol⁻¹) and 14.8 kcal mol⁻¹ (61.9 kJ mol⁻¹) below vinyl + NO at the CAS + 1 + 2 + Q level. The transition state for ring closure is predicted to provide the dominant bottleneck to bimolecular reaction for room temperature and higher. The calculated zero-pressure rate coefficient for the reaction is well fit by the modified Arrhenius expression $5.02 \times 10^{-11} (T/298)^{-3.382} \exp(-516.3/T) \text{ cm}^3 \text{ molecule}^{-1} \text{ s}^{-1}$ (T in K) for temperatures ranging from 200 to 2000 K.

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References

- M. Frenklach, D. W. Clary, W. C. Gardiner and S. E. Stein, *Proc. Combust. Inst.*, 1985, **20**, 887.
- J. H. Kiefer, *Int. J. Chem. Kinet.*, 1993, **25**, 215–219.
- A. Fahr and A. H. Laufer, *J. Phys. Chem.*, 1990, **94**, 726–729.
- V. D. Knyazev and I. R. Slagle, *J. Phys. Chem.*, 1996, **100**, 16899–16911.
- V. D. Knyazev, S. I. Stoliarov and I. R. Slagle, *Proc. Combust. Inst.*, 1996, **26**, 513–519.
- H. E. Hunziker, H. Knepe, A. D. McLean, P. Siegbahn and H. R. Wendt, *Can. J. Chem.*, 1983, **61**, 993–995.
- C. D. Pibel, A. McIlroy, C. A. Taatjes, S. Alfred, K. Patrick and J. B. Halpern, *J. Chem. Phys.*, 1999, **110**, 1841–1843.
- M. B. Pushkarsky, A. M. Mann, J. S. Yeston and C. B. Moore, *J. Chem. Phys.*, 2001, **115**, 10738–10744.
- M. Shahu, C.-H. Yang, C. D. Pibel, A. McIlroy, C. A. Taatjes and J. B. Halpern, *J. Chem. Phys.*, 2002, **116**, 8343–8352.
- A. G. Sherwood and H. E. Gunning, *J. Phys. Chem.*, 1965, **69**, 1732–1736.
- A. G. Sherwood and H. E. Gunning, *J. Am. Chem. Soc.*, 1963, **85**, 3506–3508.
- S. W. Benson, *Int. J. Chem. Kinet.*, 1994, **26**, 997–1011.
- H. Ogura, *Bull. Chem. Soc. Jpn.*, 1978, **51**, 3418–3425.
- R. Sumathi, H. M. T. Nguyen, M. T. Nguyen and J. Peeters, *J. Phys. Chem. A*, 2000, **104**, 1905–1914.
- J. S. Pilgrim, R. T. Jennings and C. A. Taatjes, *Rev. Sci. Instrum.*, 1997, **68**, 1875–1878.
- S. Yamashita, *Chem. Lett.*, 1975, **9**, 967–970.
- A. Fahr and S. E. Stein, *Proc. Combust. Inst.*, 1989, **22**, 1023.
- J. A. Seetula, in *Annales Academiæ Scientiarum Fennicæ, Series A, II. Chemica*, 1991, pp. 1–45.
- J. A. Seetula, J. J. Russell and D. Gutman, *J. Am. Chem. Soc.*, 1990, **112**, 1347–1353.
- A. D. Becke, *J. Chem. Phys.*, 1993, **98**, 5648–5652.
- W. J. Hehre, L. Radom, J. A. Pople and P. v. R. Schleyer, *Ab Initio Molecular Orbital Theory*, Wiley, New York, 1987.
- M. J. Frisch, G. W. Trucks, H. B. Schlegel, G. E. Scuseria, M. A. Robb, J. R. Cheeseman, V. G. Zakrzewski, J. A. Montgomery, Jr., R. E. Stratmann, J. C. Burant, S. Dapprich, J. M. Millam, A. D. Daniels, K. N. Kudin, M. C. Strain, O. Farkas, J. Tomasi, V. Barone, M. Cossi, R. Cammi, B. Mennucci, C. Pomelli, C. Adamo, S. Clifford, J. Ochterski, G. A. Petersson, P. Y. Ayala, Q. Cui, K. Morokuma, D. K. Malick, A. D. Rabuck, K. Raghavachari, J. B. Foresman, J. Cioslowski, J. V. Ortiz, B. B. Stefanov, G. Liu, A. Liashenko, P. Piskorz, I. Komaromi, R. Gomperts, R. L. Martin, D. J. Fox, T. Keith, M. A. Al-Laham, C. Y. Peng, A. Nanayakkara, C. Gonzalez, M. Challacombe, P. M. W. Gill, B. Johnson, W. Chen, M. W. Wong, J. L. Andres, C. Gonzalez, M. Head-Gordon, E. S. Replogle and J. A. Pople, GAUSSIAN 98, 1998.
- J. A. Pople, M. Head-Gordon and K. Raghavachari, *J. Chem. Phys.*, 1987, **87**, 5968–5975.
- J. M. L. Martin, *Chem. Phys. Lett.*, 1996, **269**, 669–678.
- T. H. Dunning, *J. Chem. Phys.*, 1989, **90**, 1007–1023.
- D. Feller and D. A. Dixon, *J. Chem. Phys.*, 2001, **115**, 3484–3496.
- T. J. Lee and P. R. Taylor, *Int. J. Quantum Chem.*, 1989, **Supplement 23**, 199–207.
- T. J. Lee, A. P. Rendell and P. R. Taylor, *J. Phys. Chem.*, 1990, **94**, 5463–5468.
- P. J. Knowles and H.-J. Werner, *Chem. Phys. Lett.*, 1988, **145**, 514–522.
- H.-J. Werner and P. J. Knowles, *J. Chem. Phys.*, 1988, **89**, 5803–5814.
- R. D. Amos, A. Bernhardsson, A. Berning, P. Celani, D. L. Cooper, M. J. O. Deegan, A. J. Dobbyn, F. Eckert, C. Hampel, G. Hetzer, P. J. Knowles, T. Korona, R. Lindh, A. W. Lloyd, S. J. McNicholas, F. R. Manby, W. Meyer, M. E. Mura, A. Nicklass, P. Palmieri, R. Pitzer, G. Rauhut, M. Schutz, U. Schumann, H. Stoll, A. J. Stone, R. Tarroni, T. Thorsteinsson and H.-J. Werner, MOLPRO, a package of *ab initio* programs designed by H.-J. Werner and P. J. Knowles 2002.1.
- Parallel Quantum Solutions (PQS) QS16-200R Parallel Quantum-Station.
- J. A. Miller, S. J. Klippenstein and S. H. Robertson, *Proc. Combust. Inst.*, 2000, **28**, 1479–1486.
- J. A. Miller, S. J. Klippenstein and S. H. Robertson, *J. Phys. Chem. A*, 2000, **104**, 7525–7536.
- J. A. Miller, S. J. Klippenstein and S. H. Robertson, *J. Phys. Chem. A*, 2000, **104**, 9806.
- D. K. Hahn, S. J. Klippenstein and J. A. Miller, *Faraday Discuss.*, 2001, **119**, 79–100.
- S. J. Klippenstein, *J. Chem. Phys.*, 1992, **96**, 367–372.
- S. J. Klippenstein, *J. Phys. Chem.*, 1994, **98**, 11459–11464.
- J. A. Miller and S. J. Klippenstein, *Int. J. Chem. Kinet.*, 2001, **33**, 654–668.
- J. D. DeSain, S. J. Klippenstein, J. A. Miller and C. A. Taatjes, *J. Phys. Chem. A*, 2003, **107**, 4415–4427.
- S. J. Klippenstein, A. F. Wagner, R. C. Dunbar, D. M. Wardlaw, S. H. Robertson and J. A. Miller, VARIFLEX 1.12m, 2002.
- J. A. Miller, S. J. Klippenstein and C. Raffy, *J. Phys. Chem. A*, 2002, **106**, 4904–4913.
- J. A. Miller and S. J. Klippenstein, *Phys. Chem. Chem. Phys.*, 2004, **6**, 1192–1202.
- J. A. Miller and S. J. Klippenstein, *J. Phys. Chem. A*, 2003, **107**, 2680–2692.